The Handoff Nobody Owns: Measuring What Cross-Mechanism Authorization Seams Cannot Carry

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Abstract

Agent authorization rarely stops at a mechanism boundary. A sender narrows its authority, a bridge translates the narrowed decision into a record the receiver can read, and the receiver applies its own admission rules. Each component can behave correctly in isolation while the composed system admits a call the sender never authorized. This paper studies that cross-mechanism handoff as a measurement problem. We describe a locked-artifact method: predictions, cases, arms, and hashes are frozen before execution; runs are sequential, offline, and deterministic; discrepancies are reconciled verbatim; and reported values are generated from stored artifacts rather than transcribed by hand. Across the mechanism pairs, the action-narrowing extension, and the boundary study of bridge variants, we test whether composition adds authorization failures beyond those a receiving mechanism already exhibits standalone. On the frozen case families, the answer is scoped: under narrowing-preserving bridge mappings, composition adds no admission beyond the receiver's own, with a measured exception in an action-narrowing family whose attribution depends on the scoring convention. The exception is bounded to its design point and case family. We distinguish receiver-side and bridge-side repair loci: consumption or evidence carriage when the narrowed fact crosses intact, and derivation repair when the bridge itself widens authority. We close with the conformance kit prepared for others to run these seam tests against their own bridge and mechanism pairs.

Keywords: cross-mechanism authorization; agent authorization; authorization seams; scope narrowing; evidence carriage; conformance testing; locked-artifact methodology

1. Introduction

At every cross-mechanism handoff in agent authorization, three parties meet at a seam that nobody owns. The sender holds authority A over a call space C and narrows that authority to $A' \subseteq A$ before sending the call onward. The bridge translates the sender's narrowed decision and state into a receiver-side record $\rho_R = B(A', \text{state})$. The receiver, running its own mechanism $M_R = (V_R, \text{admit}_R)$, decides whether to admit the call. Each party can be faithful inside its own world: the sender's narrowing can be exact, the bridge's transport can be byte-perfect, and the receiver's verdict can still be an admission the sender never authorized. The gap is not that the record is unfaithful; it is that a faithful scope record cannot carry, and the receiver cannot derive, an authorization decision. A call can cross the seam on a byte-perfect record while the authority behind it does not cross with it. That is the handoff problem this section locates: no single mechanism owns the seam, and the seam carries less context than the receiver's checks would need to detect standing authority violations on their own.

Why a faithful record is not enough is the formal core's blindness point, introduced here as a decision rule rather than as a measured result: "carried but unread = uncarried (for decision purposes)." The receiver's read set $V_read \subseteq V_R$ is the set of fields the decision actually consults — not what the record carries, but what the decision reads. A fact in the carried record that no decision reads constrains no decision, even when the transport was byte-perfect. The narrowed action can cross intact and the synthesis can simply never consume it. Measured instances of this blindness appear in Sections 4–5; in this section the aphorism does the work of the formal core's decision rule, and nothing in this section is presented as a measurement.

The research question follows directly: at a cross-mechanism handoff, does the composed system enforce the sender's narrowing, or does the seam admit calls the narrowing never conferred?

Conditional answer preview: under narrowing-preserving mappings (canonical T0) on the frozen case sets, the seam test shows zero seam — composition creates no authorization failure that one of the mechanisms does not already exhibit standalone — with one measured, convention-scoped exception (the E1 action family under arm-native scoring) that is disclosed and bounded where it is reported later, not claimed as a universal here. The zero is a recorded property of narrowing-preserving mappings on these case sets, not a theorem: as operationalized, the seam test can fire only where the composition widens, and the single measured composition-specific exception is the E1 action family under arm-native scoring, where it does. This is the bounded zero-seam finding, ⋆ `SUPPORTED`: zero seam is a property of narrowing-preserving mappings on the frozen case sets, bounded rather than assumed, and it holds only with its three conditions named together — narrowing-preserving mappings (canonical T0), frozen case sets, and disclosure of the exception and bridge artifacts. Where a bridge is not narrowing-preserving, the seam can leak with or without composition; that fault belongs to the bridge's manifest derivation rather than to composition itself, and it is disclosed here only as a condition, without numbers. Zero seam is therefore not universal in either direction: it does not assert seam safety, and it does not assert general seam failure. The star (⋆) marks the frozen cross-cutting claim, not a significance test. The zero is also partly construction-forced: in Pilot 1 and the SentinelAgent pair, the composed arm and the standalone receiving arm share the same contract and scoring, so their leak sets are identical by construction and carry no evidential weight about composition. E1 is the only genuine composition-specific test in this program; it is one design point × four attack cases and is convention-dependent. The zero is therefore a bounded design property, not an empirical discovery that composition cannot add authority.

Hypothesis key. H1 — trusted-request laundering / scope trust; H2 — dynamic state, identity, and freshness do not cross and leak; H3 — state/identity vocabulary gaps manifest as silence, not misparse; H4 — no shared audit trail (`UNTESTED`); H5 — verification attests key possession and integrity, not authority; H6 — similar-pair vs dissimilar-pair failure shapes; H7 — intent layer P2 (`UNTESTED`); ⋆ — the cross-cutting zero-under-preservation finding. The frozen experiment artifacts predate this key and use some labels differently; Appendix Table A1 records the lineage, and a legacy label is not read as the current label with the same number.

Scope-trust framing, carried as H1 `REFINED`: at a cross-mechanism handoff the receiving mechanism re-derives scope, or trusts the already-authorized caller, in a way that can admit a call the upstream narrowing did not confer. The gap is not that the record is unfaithful but that a faithful scope record cannot carry, and the receiver cannot derive, an authorization decision. Scope trust matters precisely because the receiver's re-derivation can differ from the sender's intent: a call admitted inside the receiver's own vocabulary may never have been authorized at the sender at all. The resource clause of this framing is not supported. The action clause is measured under M0; its composition-specific attribution is read under arm-native scoring. H1 is refined, not settled: the action clause is measured under M0, with the composition-specific attribution under arm-native scoring; the resource clause is not supported; and the framing is extended to the bridge's manifest derivation where that derivation widens scope. Nothing here upgrades either clause into a universal claim about scope laundering across arbitrary bridges.

Key-distribution preview, carried as H5 `REFINED`: a validly signed token attests to integrity and key possession, not the source of authority — scoped to the SentinelAgent pair; key distribution is the critical interface at the seam. A receiver that verifies a signature learns key possession and payload integrity; it does not learn who minted the record, and any keyholder is, to that receiver, equivalent to the issuer within the token vocabulary. Verification therefore attests to the token's intactness and to possession of the signing key, while the question of authority — who is entitled to act — must be settled elsewhere, at the key-distribution interface. That preview is scoped to the SentinelAgent pair and to the frozen prototype's key model; nothing here extends it to other key models or to live attestation services, and nothing here says that signed evidence is unsafe or that verification itself is the flaw.

Disclosures: conditional language only — nothing in this section is an unconditioned universal. The seam is not asserted safe by default, and no claim about seam safety in either direction is made without its conditions. Nothing here asserts priority over prior work, and nothing is framed as a new contribution beyond the frozen positioning. Every positive sentence in this section stays conditional: the zero-seam result, the scope-trust framing, and the key-distribution preview all carry their conditions in the same passage. Figure 1 presents this anatomy as a schematic.

The schematic carries no data and no numbers, and no labels beyond those defined here.

Sender: call $c \in C$; authority $A \subseteq C$; narrows $A \to A'$. Bridge (the seam) B: $\rho_R = B(A', \text{state})$ — the receiver-side record built from the narrowed decision plus state. Receiver: mechanism $M_R = (V_R, \text{admit}_R)$; read set $V_\text{read} \subseteq V_R$ — the fields the decision consults, not what the record carries. Composition: $\text{admit}_C(c) := \text{admit}_R(c, B(A'))$; effective authority $A_C := \{ c: \text{admit}_C(c) = \text{allow} \}$.

A fact in the carried record that no decision reads constrains no decision — that is the blindness the formal core

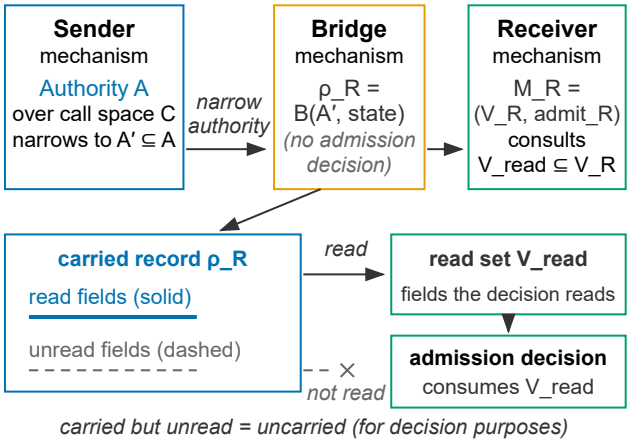


Figure 1. Anatomy of a cross-mechanism handoff — the sender narrows authority, the bridge translates it into a receiver-side record, and the receiver decides; the schematic distinguishes the record the bridge carries from the read set the receiver's decision actually consults.

names, and it is why the seam can admit what the sender never narrowed. Sender, bridge, and receiver each discharge their own duties, and no mechanism is assigned responsibility for the seam between them.

2. Background — APC (Agentic Principal Chain), TraceGrant, SentinelAgent; Progent as reference

This section places the three formal spines whose within-mechanism guarantees leave the cross-mechanism slot uncovered, and it reports each upstream guarantee as its authors' self-report — never upgraded to a manuscript result and never cited as this program's measurement. None of the three spines supplies a cross-mechanism composition guarantee; the observation is scoped to these three systems and is not a universal negative. The purpose is to show where each guarantee stops, so that the uncovered slot — the seam between mechanisms — can be named plainly.

APC [APC] — the Agentic Principal Chain mechanism — reports within-mechanism authorization: a scope algebra with meet $\sqcap$, conditions C1–C6, and Thm 4.7 (Composition Soundness), conditional on a complete effective restriction set, serialized admission, and a static X within a session. APC's "composition" is composition of action sequences inside one session; it is not composition across mechanisms. Its resource meet holds inside the mechanism. APC alone governs calls inside one mechanism; it says nothing about what happens when a narrowed call crosses into another mechanism's vocabulary, and it supplies no cross-mechanism composition guarantee. Its authorization reasoning is sound only relative to the assumptions the theorem names, and nothing in APC extends that reasoning across a bridge to a receiver with a different vocabulary.

TraceGrant [TraceGrant] reports within-mechanism delegation: POEC contracts, Eq. 2 / Algorithm 1. Its soundness is assumed (Eq. 10 is not proven). Like APC, TraceGrant's guarantee stays inside its own mechanism and extends no composition claim across a seam. Delegation soundness is an assumption of the formalism, not a result this program verifies, and it covers calls handled within TraceGrant's own contract model rather than calls handed off to another mechanism.

SentinelAgent [SentinelAgent] reports a token model τ with principles P1–P7 (P1 and P3–P7 TLA+-checked; P2 probabilistic), a TLA+ model of 2.7M states — an upstream self-report, not a measurement of this program — and Thm 4 (Composition Safety with Write-Impact Notification), which is stated for two delegation chains that both satisfy P1–P7 within its own framework. It does not state a guarantee for a chain handed to a different mechanism's decision procedure; that boundary is where this paper's cross-mechanism question begins. That figure is the authors' own model-scale claim; this program neither measured it nor relies on it as evidence.

The uncovered slot: within-mechanism guarantees do not compose across mechanisms. None of APC, TraceGrant, or SentinelAgent supplies a cross-mechanism composition guarantee. Scoped to these three — no universal negative — the slot remains: what happens at the seam between a sender's narrowing and a receiver's decision when the two mechanisms do not share a vocabulary or a decision rule. The three scopes differ in their limits, but they

share the same boundary. APC's theorem is conditional on a static X and governs intra-session action-sequence composition inside one mechanism, so it says nothing about a call crossing into another mechanism's vocabulary. TraceGrant's soundness is assumed rather than proven (Eq. 10), so its delegation guarantee covers contracts handled within TraceGrant's own model, not calls handed off to another mechanism. SentinelAgent's theorem addresses composition of two chains within its own P1–P7 framework, not a chain handed to a different mechanism's decision procedure, so the cross-mechanism case falls outside it. Each guarantee stops at its mechanism's edge: a record can be faithful at the bridge and still carry nothing the receiver's decision reads; a signature can be valid and still attest nothing about the source of authority; a theorem conditioned on a static state says nothing about a dynamic handoff.

This paper's position occupies that slot: narrowing-preservation plus the blindness lemma — cross-mechanism composition — the slot the others leave uncovered. The blindness lemma says a fact carried but never read constrains no decision; narrowing-preservation requires the composed system to enforce at least the sender's narrowing on the relevant family. Together they name the guarantee the three spines leave to the seam, and this program measures whether it holds on the frozen case sets. The claim is deliberately scoped: it concerns these three spines, these mappings, and these case sets, and it is not a universal statement about every bridge or every mechanism pair.

Progent [Progent] is a close reference point rather than an input to this program: it enforces monotonic confinement inside a single privilege-control layer, where an SMT solver classifies each policy update as a narrowing or an expansion; this paper asks whether a comparable narrowing survives translation across a mechanism boundary. Its reported results remain its authors' self-report and are not used as an input or comparison here. Nothing about Progent is claimed or measured in this section, and its results play no role in the paper's measured claims.

Disclosures: say "reports," never upgrade a self-report to an established result; upstream guarantees and numbers stay self-reported; nothing in this section impugns any upstream mechanism; no production-safety claim; no benchmark comparability; citations are pointers, not verified inputs. Citation keys are bracketed — [APC], [TraceGrant], [SentinelAgent], [Progent] — and no bibliographic detail is invented. The industry/standards citation layer belongs to Section 6 and is not invoked here. Table 1 summarizes each formal spine and the scope of its composition claim.

Table 1. Formal spine and composition-claim scope.

System	Formal spine	Composition-claim scope
APC	scope algebra, meet $\sqcap$, conditions C1–C6, Thm 4.7 (Composition Soundness)	within APC, intra-session; conditional on complete restriction set, serialized admission, static X
TraceGrant	POEC contracts, Eq. 2 / Algorithm 1	soundness assumed (Eq. 10), not proven
SentinelAgent	token τ, P1–P7, TLA+ (2.7M states), Thm 4 (Composition Safety)	within its own delegation framework; no cross-mechanism handoff guarantee
This paper	narrowing-preservation + blindness lemma	cross-mechanism composition — the slot the others leave uncovered

Table note: the SentinelAgent "2.7M states" cell is an upstream self-report, not a measurement of this program; the TLA+ model covers P1 and P3–P7, while P2 (intent) is probabilistic.

3. Method

This section presents the method as a contribution in its own right: a measurement discipline for cross-mechanism authorization seams, built so that the numbers it produces can be re-derived from locked artifacts by an independent reader. Cross-mechanism authorization invites measurement failures that are hard to detect after the fact: scoring conventions chosen once the data are visible, and counts typed by hand into reports. The method is designed against both. It combines pre-registration everywhere with a lock-before-run execution stage; it generates every reported number from run artifacts rather than typing it; it reconciles predicted and observed outcomes verbatim rather than editing them into agreement; and it submits the consolidated record to a model-assisted adversarial review and a second quality-assurance pass before anything is frozen. The section proceeds from that discipline (Section 3.1) through the formal objects it measures (Section 3.2) and the seam-test predicate (Section 3.3) to the scoring conventions that govern every classification (Section 3.4), the execution program and its ledger (Section 3.5), and the scope the method accepts (Section 3.7). Outcomes are reported nowhere in this section; where a sentence might otherwise forecast what a run found, it is phrased as what the method does or as how a quantity is defined.

3.1 Pre-registration by lock

Every experiment in the program is locked before it runs. A design artifact freezes the case set, the arm roster, the predictions the run will be scored against, and the fingerprints of the executed inputs; the lock record exists before any execution begins, and the executed cases are the locked ones. Lock records and their fingerprints — for datasets, case sets, and harnesses — are part of each run's artifact tree: for the E1 extension's dataset lock and the discriminating tests' lock, for the SentinelAgent pair's lock and case set, and for the boundary run's lock and harness. Execution is strictly sequential, offline, and deterministic. Determinism is audited rather than presumed: the protocol replays locked executions and compares them against the originals, and in the SentinelAgent run the replay covered 20 executions with 0 mismatches.

A run then passes through the stages that discipline what may be said about it:

```
lock freeze cases, arms, predictions, hashes - before any run
run strictly sequential, offline, deterministic
audit anchors: oracle reproduction, case-dict equality, field checks
reconcile predicted-versus-observed recorded verbatim; never edited in
publish every number generated from artifacts; byte-verbatim embeds
```

The audit stage anchors a run's results to checks that an independent reader can reconstruct: reproduction of the scoring oracle, equality of the run's case dictionary against the locked cases, and field-level validation of the run records. The reconciliation stage is the integrity core of the method: whenever an observed outcome differs from a locked prediction, the difference is recorded verbatim, whether or not it favors the hypothesis under test, and it is not edited into agreement afterwards. The publish stage closes the loop: numbers are not transcribed by hand into the paper — every reported number is generated from the stored artifacts, and where the prose embeds an artifact value, it embeds that value byte-verbatim. A consequence we adopt throughout: a total is written only as the explicit sum of its enumerated components, and no rounded aggregate is used in its place.

3.2 The formal core

The definitions in this subsection are carried as instruments: they state what the program scores, not what any run observed. A call space C collects calls c = (tool, resource, arguments, …). A mechanism M = (V_M, admit_M) comprises a vocabulary V_M — the fields the mechanism can represent and, structurally, read — and an admission function admit_M(c, ρ) $\in$ {allow, deny}, evaluated over a call and a receiver-side record ρ. Authority A $\subseteq$ C is the call set a principal confers; a sender narrows authority by taking the scope meet, so that A $\rightarrow$ A$'$ $\subseteq$ A. A bridge B is the seam's translation step: it consumes the sender's narrowed decision together with state and writes the receiver-side record ρ_R = B(A$'$, state). Composition is then evaluated through the bridge: composed semantics admit_C(c) := admit_R(c, B(A$'$)), and the composition's effective authority is A_C := { c: admit_C(c) = allow }. Finally, the read set V_read $\subseteq$ V_R is the set of fields the receiver's admission decision actually consults — not the fields the record carries, but the fields the decision reads.

Mapping taxonomy. The bridge variants named in the results are defined here. **M0** is the one-step, transport-faithful, scope-only bridge: it carries the sender's narrowed scope but no upstream decision. **M2** is digest-only evidence carriage; **M2-auth** is HMAC-signed evidence carriage; **M0-A** is M0 with a consuming action-scope gate. **T0** is the canonical narrowing-preserving mapping used by the primary runs; **V1** and **V2** are the two deliberately non-narrowing-preserving manifest derivations in the boundary run (Section 4.4). These labels name implementation choices, not properties of APC, TraceGrant, or SentinelAgent as published.

A bridge is narrowing-preserving, in the effective form adopted here, on a family F of cases iff the composition enforces at least the sender's narrowing there: A_C $\subseteq$ A$'$ on F. This definition admits a transport reading, concerned with what crosses the seam, and an effective reading, concerned with what the composed decisions enforce. The choice is recorded as open on the definitions, and the effective form is adopted: a narrowing that a composition does not enforce should not, under the term as used here, be counted as preserved. Both readings remain available, and any classification that depends on the choice carries the choice with it (Section 3.4). The definitions are contributed as measurement instruments, not conclusions: they say what will be scored, not what was found (Section 3.3 makes this explicit for the seam test).

3.3 The seam-test predicate

The seam test is defined case-wise over the pilot-level arms:

$$\text{seam}(case) := \text{safe}(\text{arm } 1) \wedge \text{safe}(\text{arm } 2) \wedge (\neg\text{safe}(\text{arm } 5) \vee \neg\text{safe}(\text{arm } 6) \vee \neg\text{safe}(\text{arm } 7))$$

where safe(arm) holds exactly when no call outside that arm's own world's conferred authority executes. Arm 1 carries the sender's narrowed decision, arm 2 is the receiving mechanism evaluated standalone, arm 5 is the scope-only composition, and arms 6 and 7 are the evidence and fail-closed interventions. The predicate holds when arms 1 and 2 remain within their own conferred authorities while at least one of arms 5, 6, or 7 does not. This is the pilot-level definition recorded in the artifacts; the E1 discriminating test uses the narrower arm-5 form and names that convention with its counts. Which convention scores each safe(·) term is part of the test's registration rather than a choice made after the fact: scoring conventions are explicit per test, and each run's disclosures are recorded with that run's artifacts. No outcome of the predicate is asserted in this section; its applications, with their effective sample sizes and named conventions, belong to the result sections.

3.4 Scoring conventions

Every classification of an arm's admissions names the scoring convention under which it was scored. The convention set is closed, and it contains four members: a frozen family-wide convention; an arm-native convention; a boundary convention with a canonical member and a variant-manifest member; and an arm-invariant convention for the SentinelAgent pair. Table 2 states what each convention scores, at definition level.

Table 2. Scoring-convention matrix — the four classification conventions and what each scores (definitional rows; no run data).

Convention	What it scores
Family-wide (frozen)	Scores each arm of a mechanism family against a shared authorization norm derived from the handoff, including arms whose own world contains no handoff
Arm-native	Scores each arm against its own world's conferred authority
Boundary canonical / variant-manifest	Scores the standalone arm canonically regardless of the surrounding variant loop variable, and scores the variant arms against the authority that their variant-manifest derivation produces
SA arm-invariant	Scores by an admission relation applied identically across all arms of the family, including the standalone arm (role-resource conjunction, with the prohibited-composition rule applied to the standalone arm as well)

Conventions are carried by disclosure rather than by report-time choice. Where more than one convention can classify the same records, each reported classification states the convention that produced it, and any selection rule between readings is stated with the report rather than assumed. This requirement binds the result sections: every count, contrast, or reconciliation that those sections report carries its convention, so that no number can be read outside the rule that generated it.

3.5 The execution ledger

The matrix execution ledger is an explicit record, transcribed from the frozen evidence base; Table 3 states it as an exact sum of its component runs, in the form the publish stage requires. The ledger includes E1's six control executions in the `+ 6` component: they are inside the 2,446 total. The SentinelAgent run's 10 control executions and the boundary run's 7 control executions are outside the total, as are the determinism replays and the separate Pilot-1 control, replay, and latency records; these secondary records are listed in the note below Table 3 so that no execution is counted twice. Control executions pin the admission behaviors on their own records.

Table 3. Matrix execution ledger — the program's matrix executions, stated as an explicit sum of its enumerated components.

Run	Design	Executions
Pilot 1 (v2) — APC → TraceGrant	80 cases × 7 arms	560
Extension E1 — action-narrowing + M2-auth	8 cases × 5 arms	40 + 6 control executions
Discriminating Test 1 (re-scoring only)	no new executions	0
Discriminating Test 2	8 cases (4 attack + 4 benign) × 2 arms	16
SentinelAgent pair — APC → SentinelAgent	96 cases × 7 arms	672
Boundary — manifest-derivation variants (V1/V2)	6 arms × 2 variants × 96 cases	1,152
Total		**2,446 = 560 + 40 + 6 + 16 + 672 + 1,152**

Path notation. The four artifact roots are:

- `v2` = `tests\seam_pilot_v2\results`
- `dt` = `tests\seam_pilot_v2\discriminating_tests\results`
- `sa` = `tests\seam_pilot_sentinelagent\results`
- `b` = `tests\seam_pilot_sa_boundary\results\boundary`

Full paths remain in the artifact tree.

Outside the scale column (excluded from the total). The SentinelAgent determinism replay, 20 executions; SentinelAgent control executions, 10; boundary control executions, 7; Pilot-1 control battery, 21 records; Pilot-1 determinism replay, 10 pairs; and the seven-arm latency harness, 100 warmup + 1,000 measured samples per arm. The latency harness output is `v2\latency.json`. E1's six control executions are inside the total, carried in the E1 row's `+ 6`. Totals are stated as explicit sums of enumerated components; no rounded aggregate is used.

Implementation provenance. APC's sender is the pinned upstream `apc/` package (commit `d31a1ea115a3`, abbreviated). The SentinelAgent receiver is the pinned upstream prototype (`sentinelagent_das_prototype.py`, SHA-256 `739d9db1e836`, abbreviated), with only the module-level key constant set by the harness. The TraceGrant receiver is the independent Task D implementation under `tests\tracegrant_task_d_20261014\implementation\`, wrapped without modification; its contract mapping is this program's design choice, not a reading of either paper. The bridge mappings throughout are this program's constructions, not upstream translations.

3.6 Review loop

The lock–run–publish discipline is completed by a review loop. After runs, the claim records are consolidated; the consolidated record is examined by a model-assisted adversarial review and again by a second quality-assurance pass, and the corrections raised are applied at source before the record is frozen. Reconciliation within this loop treats every predicted-versus-observed discrepancy verbatim, with no silent normalization. Where the results sections name upstream work — APC, TraceGrant, SentinelAgent, or Progent — it is reported as its authors state it, not re-derived here.

3.7 Ethics, AI use, and reflexivity

AI-assisted drafting, review, and quality assurance were used throughout this work. The SentinelAgent adversarial review was performed by `GLM-5.3-Flash Free`; a fresh-context OpenCode cross-check and a targeted external free-chat review were also used. These are not human peer review and do not constitute independent vendor validation. No AI system is an author; the authors remain responsible for all claims, artifacts, and errors. No human-subjects data were collected; the work is synthetic and offline.

4. Results

This section reports three pilots and a boundary run. Each subsection states, above its tables, the classification convention under which its classifications are scored and the effective witness set behind every count. All tables are

generated from the frozen result artifacts and transcribe the artifact fields verbatim; the source line under each table names the file and fields from which every cell is drawn.

4.1 Pilot 1 — APC → TraceGrant (v2)

Pilot 1 is the first end-to-end pair of the program: a sender running APC narrows its authority and hands off to a receiver running TraceGrant. The pilot's purpose is to separate what crosses the seam cleanly from what does not, on 80 cases across seven arms — 40 attack and 40 benign cases per arm — with classifications scored under the frozen family-wide convention: the same reference rules applied to every arm on both sides of the handoff. Effective witness sets: the 40 attack cases are 10 templates × 4 attack parameters (8 runs each, attack and benign interleaved); and the 24 leak cases of the headline arm are 6 leak templates × 4 parameters — one design point per row, not 24 independent demonstrations. The pilot is a constructed case family rather than a benchmark, and no comparability with published benchmark numbers is claimed.

The division of labor between the two mechanisms is narrowed before the handoff, so the pair separates cases where a checkable fact exists on the receiving side from cases where no faithful record could make the fact checkable. What is checkable is denied. The four `resource_widen` attacks are dropped in every arm — including both standalone mechanisms (the sender and the receiver) — because the sender's scope meet holds on the resource side and the narrowed set, once crossed, lets any downstream check see the widened call for what it is. The resource clause of the scope-trust claim (H1 `REFINED`) is therefore **not supported**: trusting the upstream record on resources does not admit anything the upstream did not confer. The two representable attack classes inside the shared vocabulary — recipient substitution and the omitted-forbidden default — are caught (leaked 0) by mapping-side canonicalization and BIND constraints rather than by effect-time detection. They are caught-case and mapping-attribution findings, not silence instances; they do not support H3, and they are reported under H1 `REFINED` and the measured findings. The one seeded exception is the unsatisfiable-VERIFY template, treated as a completion-integrity boundary rather than an unauthorized-effect leak and also not an H3 instance.

What leaks is the unrepresentable. Six of the ten templates leak 4 cases each under the composed M0 arm — principal rebind, composition-history loss, repeated calls (4 cases but 6 unauthorized-effect events), budget reset, expired delegation, revoked envelope — for a total of 24 of 40 attack cases and 26 unauthorized-effect events, with confidential marker external effects recorded in 4 of those events. These are exactly the fields of dynamic state, identity and freshness that cannot be written as a checkable field inside the receiver's vocabulary at all: no decision on the receiving side reads them, so the same faithful transport that delivers resource scope correctly delivers nothing actionable about revocation or budget. Template-level leak counts and event counts are given in Table 5. Because the classes are aligned row-by-row with the six templates, the composition-type claim (H2 `SUPPORTED`) — that dynamic state, identity and freshness do not cross and leak — is carried here with every count attached to its row; its cross-pair headline appears only in Section 4.3 and is not invoked in this section. Under the frozen family-wide convention, no Progent figure is used for comparison here; H6 `SUPPORTED` rests on the two measured receiving pairs, not on an external baseline.

Table 4 summarizes the per-arm outcomes.

Table 4. Pilot-1 arm outcomes (attack cases leaked, unauthorized-effect events, and benign false-denial cases per arm).

Arm	Attack cases leaked	Unauthorized-effect events	Benign false denial cases
`apc_alone`	0	0	0
`tracegrant_alone`	24	26	0
`apc_apc_preserved`	0	0	0
`apc_apc_reset`	8	8	0
`apc_tracegrant_scope_only`	24	26	0
`apc_tracegrant_retained`	0	0	0
`apc_tracegrant_fail_closed`	0	0	20

Reset-control note: `apc_apc_reset` is the frozen same-mechanism reset control: 8 leak cases / 8 events on the shared 80 (composition 4, repeated 2, budget 2), identical to the frozen run's arm-4 cells.

Receipt-failure note. The standalone receiver recorded 4 receipt failures. `v2\raw_metrics.json` records `tracegrant_alone.receipt_failures = 4`; this control detail is not part of the 24/40 leak count and is not load-bearing for H2.

M0 is this program's one-step, transport-faithful bridge — a design choice of this implementation and not a defect of APC or of TraceGrant as published — and the composed arm with M0 (`apc_tracegrant_scope_only`) exhibits leak counts and event counts identical row-by-row to the standalone receiver in Table 5: the handoff does not add a dynamic-state leak class, it fails to remove the receiver's own. The informative rows are the remaining arm comparisons. The arm that carries the sender's decision as an evidence record with a consistency digest (`apc_tracegrant_retained`, M2) shows 0 leaked attack cases and no false denials on the same case set. M2 did not sign the record: it was trusted harness input with a consistency digest. HMAC-signed M2-auth is introduced only in E1, and signature verification under a trusted key belongs to E1 M2-auth and to the SentinelAgent pair's arm 6. The 0/40 result therefore measures carriage of the upstream decision, not signature verification. The arm that instead refuses to translate what it cannot represent (`apc_tracegrant_fail_closed`) also shows 0 leaked cases, at a measured utility cost: 20 of 40 benign cases are falsely denied, in 28 distinct events.

Table 5 gives the leak-class structure.

Table 5. Leak-class table for the Pilot-1 family-wide arms (per-template leaked cases, unauthorized-effect events, and false-denial events, all under the frozen family-wide scoring convention).

Template	`tracegrant_alone` leaked	`tracegrant_alone` unauthorized events	`tracegrant_alone` false denials	`apc_tracegrant_scope_only` leaked	`apc_tracegrant_scope_only` unauthorized events	`apc_tracegrant_scope_only` false denials	`apc_tracegrant_retained` leaked
`resource_widen`	0	0	0	0	0	0	0
`principal_rebind`	4	4	0	4	4	0	0
`composition_history_loss`	4	4	0	4	4	0	0
`repeated_calls`	4	6	0	4	6	0	0
`budget_reset`	4	4	0	4	4	0	0
`expired_delegation`	4	4	0	4	4	0	0
`revoked_envelope`	4	4	0	4	4	0	0
`recipient_substitution`	0	0	0	0	0	0	0
`omitted_forbidden_default`	0	0	0	0	0	0	0
`unsatisfiable_verify`	0	0	0	0	0	0	0

The state/identity rows in Table 5 are the pilot's H3 observation: silent vocabulary/state gaps, not crashes or misparses — the receiver admits these calls with no parse error, no gate code, and no effect-time detection, because the fact that should have been denied has no column in the receiver's decision. The resource and caught-action rows are not H3 evidence.

Worked numeric trace (Pilot 1; existing numbers only). The ledger entry is 80 cases × 7 arms = 560 executions, with 40 attack cases per arm. Six leak templates × 4 attack parameters give the headline 24/40 leak cases; the per-template unauthorized-effect events are 4 + 4 + 6 + 4 + 4 + 4 = 26. The evidence-carriage arm records 0/40 leaks, and the fail-closed arm records 0 leaks with 20/40 benign false-denial cases and 28 false-denial events. Each value is the value already reported in this subsection and Tables 4–5; the trace introduces no new measurement.

4.2 Extension E1 and the resolving tests — the action family

Pilot 1 severe cases all sit on the state/identity side, so a second experiment asks the same question of a fact the receiver's vocabulary demonstrably does represent: the action. Extension E1 isolates one design point — a `send_external` call that is inside the sender's root set but outside the narrowed action set that crossed the seam — replicated across four attack cases on an axis the construction neutralizes (recipient value, budget); the four benign cases are context, and the effective sample size is 1 design point × 4 attack cases. The experiment spans 8 cases × 5 arms = 40 executions plus 6 control executions on a separate denominator (the 6 are counted in the 2,446 ledger but excluded from the 40-case E1 denominator), scored under the frozen family-wide reference convention. The reference oracle for this family is family-wide: it scores arm 2, which contains no handoff, against the handoff-derived norm — an authority-assignment convention for the standalone arm that is disclosed rather than hidden, and one that the first resolving test in this subsection re-scores.

The result decomposes transport from admission at the action field. The narrowed action set crosses the seam intact — the synthesized record in the composed arm carries exactly the narrowed set, `[read_public, save_internal]` — and the composed arm still admits the four attacks, 4/4, with 4 unauthorized-effect events; the record's action field is present in the snapshot and unread by the synthesis. This is the action-clause instance of the scope-trust claim (H1 `REFINED`), measured here: an authorization-carrying fact crosses the seam and a secondary receiving decision never reads it. In the frozen family-wide scoring the same 4/4 leaks appear in arm 2, which contains no handoff; the two M0 arms are scored identically and leak identically. The outcome matrix is given in Table 6; the four attack cases are `action_narrow-0-a` through `action_narrow-3-a`, and the four benign cases are `action_narrow-0-b` through `action_narrow-3-b` (arm mapping and case ids are also recorded in `v2\extension_e1\assertions.json` and `tests\seam_pilot_v2\design\extension_e1_cases.json`).

Table 6. E1 outcome matrix (per-arm attack leaks, unauthorized-effect events, benign false denials, and arm-6 signed-record verification on the 4 attack and 4 benign cases). Arm identifiers: arm 1 = `apc_alone`; arm 2 = `tracegrant_alone`; arm 5 = `apc_tracegrant_scope_only`; arm 6 = `apc_tracegrant_retained`; arm 7 = `apc_tracegrant_fail_closed`.

Arm	Attack leaks	Unauthorized-effect events	Benign false denials	Arm-6 verified (8)
1	0/4	0	0/4	n/a
2	4/4	4	0/4	n/a
5	4/4	4	0/4	n/a
6	0/4	0	0/4	`PROOF-GAP` after verify; 8/8 `EVIDENCE-VERIFIED` (report-only)
7	0/4	0	0/4	n/a

Three mechanism-level facts sit alongside Table 6. First, the arm-5 confusion matrix on the eight-case family is TP `4`, FP `0`, FN `0`, TN `4` — precision `1.0`, recall `1.0` under this experiment's own contract, confirming the E1 construction detects the action-family leak exactly at the composed arm without dropping benign traffic. This is a representable-but-unread consumption failure, reported under H1 `REFINED` and the formal core, not H3. Second, the evidence arm (`apc_tracegrant_retained`) denies all four attacks (`PROOF-GAP` after the shadow APC denies) while verifying the signed record on all four attack cases; the eight signed records across arm-6 runs — four attack and four benign — are all `EVIDENCE-VERIFIED`. This cell is report-only in the frozen metrics output — the raw field traces for signed-record admission codes are not stored in `v2\extension_e1\metrics.json`, and the count is carried from the frozen report's evidence-auth column (`v2\EXTENSION_E1_RESULTS.md` Section 3.4). Third, the three controls — `C-valid`, `C-mutated`, `C-replay` — each with a failure probe, pass 3/3 across 6 executions on their own denominator, all `lock_matches` true; they are not mixed into the 40-case run. The signing scheme is documented as HMAC-SHA256 over a canonical payload with a 56-byte key and is disclosed as a public test constant; the M2-auth arm assumes attestation is verified under the trusted key and does not establish authenticated transport. The dataset hash of the locked 8-case set is recorded in the E1 manifest as `1b7c6083ef22d40c8a0d3c3705357358b3508671bfbedc9db12f69a8b2e264ba` (design locks, not reseals).

Two pre-registered resolving tests then settle, respectively, where the action-clause violation is located and how the violation is repaired. Test 1 is scoring-only — zero new executions — and re-scores the four frozen arm-2 attack records against two authority conventions: the frozen family-wide handoff norm, under which they remain leaks, and the arm-native root norm, the authority arm 2's own world confers, under which arm 2 leaks 0/4. On the same frozen records the seam test — composed-arm-only admission — counts 0 under the canonical convention and 4/4 under the arm-native convention. The classification is therefore measured to be convention-dependent, and the convention-dependence is itself the finding: the composed-violation reading rests on the arm-native scoring of arm 2, which is the reading both resolving tests adopt going forward. Under that reading the action-family exception to the pair-similarity finding (H6 `SUPPORTED`) is composition-specific: on this family, this arm, this mechanism — arm 1 denies, arm 2 is safe, arm 5 leaks, with no claim about other families. The seam test as operationalized here is a widening test; it can fire only where the composition widens, and a zero where the composition does not widen does not by itself establish composition-independence. Table 7 reports both tests.

Table 7. Resolving-test matrix for the E1 action family (scoring-sensitivity panel, Test 1; consumption-repair panel, Test 2). Panel labels: T1 = Test 1 (scoring sensitivity); T2 = Test 2 (consumption repair). T1 values are out of the 4 attack cases, shown as numerator/denominator.

Panel	Test	Measured value
Scoring sensitivity (T1)	Arm-2 attack leaks under handoff norm (canonical)	4/4
Scoring sensitivity (T1)	Arm-2 attack leaks under root norm (arm-native)	0/4
Scoring sensitivity (T1)	Seam-test count under canonical scoring	0/4
Scoring sensitivity (T1)	Seam-test count under arm-native (root-norm arm-2) scoring	4/4
Consumption repair (T2)	Arm-5 attacks denied on action scope (`ACTION-SCOPE`)	4/4
Consumption repair (T2)	Arm-5 benign cases allowed	4/4
Consumption repair (T2)	Admitted calls with verifying receipt	12/12
Consumption repair (T2)	Arm-2 gate-outcome parity vs. canonical	8/8
Consumption repair (T2)	Arm-2 contract-digest parity vs. canonical	8/8
Consumption repair (T2)	Locked predictions matched (Test 1 + Test 2)	10/10

Test 2 measures the repair (M0-A consumption rule): making the synthesized read set consume `snapshot.actions` denies all 4 attacks on action scope in the composed arm while allowing all 4 benign cases, with 12/12 of the admitted-call receipts verifying across the loop and 0 false denials anywhere, all under 16 sequential executions. The arm-2 control in the same loop — the same variant, cases scored against the same contract — is unchanged: gate outcome parity 8/8 with canonical, and the synthesized contract digests byte-identical on all 8 cases, so the variant changes admission handling only, not the M0 arm's own decisions. All 10 pre-registered predictions across the two tests match; the lock hash of the pre-registration is recorded in the artifact directory as `8cd4d9938348b43c7232eaa022280693d7e6f538c630d47f0c6d63bd996011ab`. This is the measured consumption-side instance of the formal core's repair rule (Section 3.2): the receiver must consume the narrowed action set it is handed; it is reported as measured, with no claim-ladder token, and it is not imported from a claim entry in this section.

Test 2 runner-hash disclosure. `dt\test2_manifest.json` records `runner_sha256_matches_mini_lock: false`. The earliest full run crashed while assembling the reconciliation after writing the test artifacts; the corrected runner re-ran the complete matrix. The mini-lock, cases, arms, and predictions were unchanged, and the manifest's `bookkeeping_note` records the deviation; no measurement or reported count changes.

Read together, Pilot 1 and the E1 resolving tests measure two loci. At the state/identity locus, the vocabulary gap is absolute and the leak count is the standalone count. At the action locus under the frozen family-wide convention, a composed arm leaks a call its snapshot record does not authorize; under the arm-native convention the same records show the composed arm as the only violating arm, and consumption — the receiver reading the fact it was already handed — closes the violation without disturbing the rest of the synthesis. Whether the action-family exception generalizes beyond this one family, and what other loci the second and third pairs expose, is deferred to Sections 4.3–4.5.

4.3 SentinelAgent pair — APC → SentinelAgent

The second receiving pair re-runs the composition program against SentinelAgent's delegation-token vocabulary: the locked bridge mints one delegation token per attempted call, and the receiving side executes the pinned prototype's own token checks; every classification in this subsection names its convention, which here is the pair's arm-invariant scoring: scopes are intersected with the receiving manifest and the composition term is applied to arm 2 as well, although arm 2's world contains no handoff; under a rescoping of the composition term to the handoff arms alone, the arm-2 `composition_history_loss` cells would read safe rather than leak, so the pair's arm-2 and arm-5 readings are one convention's readings. The witness structure is 12 templates × 8 runs, executed as 96 cases × 7 arms = 672 strictly sequential, offline, deterministic runs over 48 attack cases. The self-review re-verified the lock and case-set hashes unchanged after the run and re-measured determinism with 20 re-executions and zero mismatches.

Under the locked convention the standalone receiving arm and the composed scope-only arm leak the same 28 of the 48 attack cases and emit the same 30 unauthorized-effect events. Their leak sets are identical by construction — same contract, same scoring, composition changing only which vocabulary handles the prelude calls — and we carry the identity as forced, not as evidence. The dynamic-state classes of the first pair repeat and dominate: 24 of the 28 leaks are principal/task identity (4 cases), composition history (4), repeated calls (4 cases with 6 events), cumulative budget (4), expiry (4), and revocation (4). This is the structural account for H3 `SUPPORTED`: the

state/identity leaking classes are fields the receiving vocabulary cannot represent, so nothing fails loudly — silent vocabulary/state gaps, not crashes or misparses. The remaining 4 signature-family leaks are not H3 evidence; they are treated under H5/H6 below. The seam test again returns zero under the locked conventions — no case is safe in both mechanism-alone arms and admitted under composition in this pair. The run's own integrity record is not uniformly green: the SentinelAgent manifest records `expected_vs_observed_ok = false`, and its metrics record `comparison.overall_ok = false`. The two failed checks are the false-denial-event lock (26 locked, 28 observed) and the arm-5 confusion true-negative denominator (68 locked, 20 observed on the attack-case basis); both are recorded verbatim in `sa\sa\reconciliation.json` with `affects_behavior_cells: false`, and Section 7.2 gives the full reconciliation. Neither changes a behavior cell or a reported leak count.

The two remediations carry measurable prices, and the pair headline is the frozen H2 claim: the repair is carrying the upstream decision, not a check inside the receiver's vocabulary (H2 `SUPPORTED`). In this pair the evidence arm records zero attack successes and zero false denials against the scope-only arm's 28 leaks. The fail-closed arm also records zero leaks, by refusing dynamic state it cannot translate, at a measured utility cost: 20 benign cases rejected and 28 false-denial events. The trust conditions are part of the finding: the evidence arm reads the signed attestation as verified under the trusted key, and the fail-closed arm accepts its 20 benign refusals deliberately — both scoped design choices of the mapping, neither claimed production-safe. H6 `SUPPORTED` reads the same evidence pair-wise: the leak shape of the two receiving mechanisms is nearly identical, and the differences that exist are remediation mechanics — signed attestation here, digest-only evidence carriage and receipts in the first pair — plus the one new standalone class the token ecosystem contributes, described below. Table 8 gives the per-template observations behind this paragraph.

Table 8. SentinelAgent pair, per-template outcome cells for the four arms that consume the token vocabulary (arm 2 `sentinelagent_alone`, arm 5 `apc_sentinelagent_scope_only`, arm 6 `apc_sentinelagent_retained`, arm 7 `apc_sentinelagent_fail_closed`): leaked cases, false-denial counts (`false_denials`), leak events. The arm-7 `false_denials` cells are per-template artifact values and are non-additive: they sum to 26, while the arm-level `false_denial_events` field is 28. The two additional events are the second-call `repeated_calls` events recorded in the reconciliation; the cells are artifact-faithful and do not sum to the arm-level total. Columns are arm number plus metric: leak = leaked cases; FD = false-denial count; ev = leak events; * marks the non-additive arm-7 FD count. Template identifiers are shown with spaces in place of underscores.

template	2 leak	2 FD	2 ev	5 leak	5 FD	5 ev	6 leak	6 FD	6 ev	7 leak	7 FD*	7 ev
resource widen	0	0	0	0	0	0	0	0	0	0	0	0
principal rebind	4	0	4	4	0	4	0	0	0	0	0	0
composition history loss	4	0	4	4	0	4	0	0	0	0	4	0
repeated calls	4	0	6	4	0	6	0	0	0	0	6	0
budget reset	4	0	4	4	0	4	0	0	0	0	4	0
expired delegation	4	0	4	4	0	4	0	0	0	0	4	0
revoked envelope	4	0	4	4	0	4	0	0	0	0	4	0
recipient substitution	0	0	0	0	0	0	0	0	0	0	0	0
omitted forbidden default	0	0	0	0	0	0	0	0	0	0	0	0
unsatisfiable verify	0	0	0	0	0	0	0	0	0	0	4	0
foreign key signed	0	0	0	0	0	0	0	0	0	0	0	0
keyholder widen	4	0	4	4	0	4	0	0	0	0	0	0

Reconciliation note (SentinelAgent, arm 7). The verbatim fields in `sa\sa\reconciliation.json` for item `arm7_false_denial_events` are: `locked`: 26; `observed`: 28; `locked_note`: "FD events = 20 benign dynamic refusals + 4 unsatisfiable_verify-a + 2 oracle-clean first calls of repeated_calls-a"; `explanation`: "The lock's derivation enumerated only 'first calls' of repeated_calls-0-a and -3-a (2 events), but each case produces two FD events: under the locked FD definition (oracle-clean AND denied AND not a signature rejection) the second call is also an FD because the refused first call never executes, leaving the oracle budget unconsumed; the shadow APC then denies the second call (SHADOW-DENY) after charging its own budget on the first approval. Observed events: repeated_calls-0-a ev0 T1-REFUSE:dynamic_state_untranslated / ev1 SHADOW-DENY; repeated_calls-3-a identical. The locked definition yields 28; only the derivation undercounted."; `seam_effect`: "none"; `affects_behavior_cells`: false.

Reconciliation note (SentinelAgent, arm 5). The verbatim fields in `sa\sa\reconciliation.json` for item `arm5_confusion_true_negative_denominator` are: `locked`: 68; `observed_attack_case_denominator`: 20;

`observed_all_cases_denominator`: 68; `explanation`: "The lock's confusion block used the all-96-cases denominator for TN (48 benign + 20 caught attacks), while the metrics code computes TP/FP/FN/TN over the 48 attack cases (TN = 16 rejected + 4 allowed-no-effect). TP/FP/FN are 28/0/0 either way; counts agree: 28+20=48, 28+68=96. Denominator slip in the lock's block, recorded, not edited."; `seam_effect`: "none"; `affects_behavior_cells`: false.

The signature family is this pair's measurement of H5, and its first end-to-end test. Its witness structure is 2 pairs × 4 document parameters × 2 variants = 16 cases (8 attack, 8 benign), each executed across the 4 token arms. A token whose content would pass the checks but which is signed with the prototype's published key constant instead of the trusted harness key is rejected `SIGNATURE_INVALID` — 4 rejections per token arm, and not counted as a false denial under the locked scoring rule. Verification without the key cannot admit; verification with the key attests something narrower than authority. Delegate-minted tokens are admitted exactly like bridge tokens wherever the prior authorization is not restored, and a control presenting identical tokens attributed to two distinct minters returns identical codes. Within the token vocabulary, any keyholder is to the receiver equivalent to the issuer, and the frozen wording of H5 `REFINED` states the measured boundary for this pair: a validly-signed token attests to integrity and key possession, not source of authority — scoped to this pair, with the trust-boundary caveat that the pair exercises one prototype, one key model, one case family, and the "or a call to the DAS" alternative was not exercised end-to-end. The same property generates the pair's one new leak class: `keyholder_widen`, in which a caller-held delegate writes a manifest containing the widened resource and the token arms admit it 4/4 — a token-ecosystem standalone class (H6 `SUPPORTED`) rather than a composition failure, and the observation that makes key distribution the critical interface of this seam.

Ten controls pass across the family — signature-validity failure probes, foreign-key rejection, two-minter indistinguishability, attestation tamper and replay — showing the checks are neither vacuous in the admit direction nor no-ops in the deny direction. Across the 7-arm, 96-case matrix (756 evaluated events), the non-vacuity count is 488 admissions plus 150 denials with a token event attached (638 counted events; 40 native P6/P7 denials, 42 attestation-stage denials in arm 6, 68 pre-presentation refusals in arm 7). The 230 admitted events without a token key are included in the 488; the 118 denials without a token key are outside the 638 count. Table 9 reports the per-template codes; the ten controls follow.

Table 9. SentinelAgent signature family - per-template codes and leaks in the four token arms. Arm 2 = standalone; arm 5 = scope-only; arm 6 = retained signed attestation; arm 7 = fail-closed shadow deny. Cells are code x4 / leaked cases; `SIG-INVALID` abbreviates `SIGNATURE_INVALID`.

template	kind	arm 2	arm 5	arm 6	arm 7
foreign_key_signed	attack	SIG-INVALID x4/0	SIG-INVALID x4/0	SIG-INVALID x4/0	SIG-INVALID x4/0
trusted_key_signed	benign	ADMITTED x4/0	ADMITTED x4/0	ADMITTED x4/0	ADMITTED x4/0
keyholder_widen	attack	ADMITTED x4/4	ADMITTED x4/4	APC-DENY x4/0	SHADOW-DENY x4/0
keyholder_plain	benign	ADMITTED x4/0	ADMITTED x4/0	ADMITTED x4/0	ADMITTED x4/0

The ten controls, with expected and observed results:

- `S1-valid` — expected p6=ALLOWED by scope; p7=OUTPUT_ALLOWED; observed p6=ALLOWED by scope:read_confidential; p7=OUTPUT_ALLOWED.
- `S1-failure` — expected p6=SIGNATURE_INVALID; p7=SIGNATURE_INVALID; observed the same.
- `S1-probe` — expected p6=TOKEN_REVOKED; p7=TOKEN_REVOKED; observed the same.
- `S2-foreign` — expected p6=SIGNATURE_INVALID; p7=SIGNATURE_INVALID; observed the same.
- `S2-trusted` — expected p6=ALLOWED by scope; p7=OUTPUT_ALLOWED; observed p6=ALLOWED by scope:read_confidential; p7=OUTPUT_ALLOWED.
- `S2-two-minters` — expected p6_both=ALLOWED by scope; p7_both=OUTPUT_ALLOWED; codes_identical=true; observed bridge p6_bridge=ALLOWED by scope:read_confidential; p7_bridge=OUTPUT_ALLOWED; delegate p6_delegate=ALLOWED by scope:read_confidential; p7_delegate=OUTPUT_ALLOWED; codes_identical=true.
- `S2-mutated` — expected p6=SIGNATURE_INVALID; p7=SIGNATURE_INVALID; observed the same.
- `S3-valid` — expected gate=ALLOWED; tamper=ATT-SIG-BAD; observed gate=ATT-OK; tamper=ATT-SIG-BAD.
- `S3-mutated` — expected gate=ATT-SIG-BAD; probe=APC-DENY; observed the same.
- `S3-replay` — expected gate=ATT-BIND-MISMATCH; probe=ALLOWED; observed gate=ATT-BIND-MISMATCH; probe=ATT-OK.

ATT-SIG-BAD = ATTESTATION-SIGNATURE-INVALID; ATT-BIND-MISMATCH = ATTESTATION-BINDING-MISMATCH; ATT-OK = ATTESTATION-OK, the attestation stage's allow code (the `ALLOWED` entries in the expected column denote the same outcome).

4.4 Boundary — non-narrowing-preserving bridge variants (V1/V2)

The boundary run asks the reversed question: not whether a faithful bridge drops facts, but what happens when the bridge's own manifest derivation is not narrowing-preserving. Two derivations answer it. V1 derives the receiving manifest from the delegate's pre-narrowing role scope rather than the post-handoff effective scope; V2 fills an omitted-recipient default into the manifest's SEND endpoint (`forbidden-default@example.test`). Neither derivation consumes a fact the sender conferred; both widen. Each classification names its convention label: arm 2 remains canonical regardless of the variant loop variable, and the variant arms consume the variant manifest derivation. The run executes 6 arms × 2 variants × 96 cases — 1,152 strictly sequential runs, 96 rows per arm-variant cell; 7 controls pass, and the self-review records `ok` with the harness and case-set hashes unchanged, while the boundary lock hash is recorded in `manifest.json.lock`. Its attack structure is deliberately thin: 1 design point × 4 cases per variant — the four `resource_widen` attacks under V1, the four `omitted_forbidden_default` attacks under V2 — so its 8 attack cases in total are not 8 independent designs.

Both derivations produce the predicted widening, with and without composition. Under V1 the composed arm leaks the 4 `resource_widen` attacks, among them the widened resource `docs:external/secret`, which the canonical arm 2 blocks; under V2 the composed arm leaks the 4 `omitted_forbidden_default` attacks. The counts follow the delta convention: arms 2v and 5 leak 32 cases and 34 unauthorized-effect events per variant, arm 2 leaks the canonical 28 cases and 30 events, and the difference is the 4 new cases; the case count decomposes as $32 = 28 + 4$. Arm 1 leaks none in either variant. The seam counts follow the same shape: 4 per variant. Because the standalone variant arm 2v leaks the identical set per variant — forced by construction, not evidence — the seam test fires, but its firing carries no composition-specific witness. The informative contrasts are canonical arm 2, safe on the variant cases, against composed arm 5, leaking them; and arm 2v, leaking, against arm 2, safe.

The attribution is measured, in the frozen wording: the failure is the bridge's manifest derivation, not composition. The receiver cannot recover a fact the derivation did not narrow — the frozen derivation-side clause. The claim readings move accordingly. H1 `REFINED`: the bridge's manifest derivation extends scope trust — the receiver reads the signed manifest as authority — and the refinement lands on the derivation layer. H2 `SUPPORTED` is unchanged: V1 and V2 add no new dynamic-state class, and their new leaks are scope/manifest-derivation classes. These are derivation-widening findings under H1 `REFINED` and H6 `SUPPORTED`, not H3: both mismatches are faithful to what the bridge wrote into the manifest — a role scope exceeding the root grant under V1, a default-filled send endpoint under V2. H5 `REFINED` now covers the keyholder property at the bridge/manifest layer: a bridge holding the key can write a widened manifest or a default-filled endpoint, and the receiver admits what it verifies. H6 `SUPPORTED`: a bridge-design artifact, not a composition-specific failure.

Arms 6 and 7 suppress the widened calls rather than repair them. The evidence arm denies the 4 variant cases as `APC-DENY` per variant, with zero leaks; the fail-closed arm denies them as `SHADOW-DENY` ×4 per variant, with zero leaks and its measured cost of 20 benign false denials and 28 false-denial events. No benign case is falsely denied in arms 1, 2, 2v, 5 or 6, and the no-divergence control (CV1b) confirms the variant manifests match canonical on the benign cases that should not diverge. The missing step is not receiver consumption here; the derivation is at fault.

The canonical zero seam survives, bounded. It is a property of narrowing-preserving mappings (canonical T0) on the frozen case sets and does not extend to non-narrowing-preserving derivations (Section 3.2). The run's manifest check is narrow, recorded verbatim: `expected_vs_observed_ok = true` checks the seam count only (Table 11 note). One locked statement was overbroad and is reconciled rather than edited (Table 10 note): arm 2 was stated as safe on all cases, and it is safe on the variant-specific cases per variant while carrying the pre-existing canonical leak set elsewhere. Finally, V1 and V2 are two derivations tested in one pair's token vocabulary; they are not asserted to be typical, nor the only or most likely bridges. Table 10 gives the absolute-plus-delta matrix and Table 11 the seam counts.

Table 10. Boundary absolute-plus-delta matrix — per arm-variant cell: leak cases, leak events, benign false-denial cases, false-denial events, and variant-template leak cases (the `resource_widen` template under V1, the `omitted_forbidden_default` template under V2).

arm::variant	leak cases	leak events	benign FD cases	FD events	variant-template leak cases
apc_alone::v1	0	0	0	0	0
apc_alone::v2	0	0	0	0	0
sentinelagent_alone::v1	28	30	0	0	0
sentinelagent_alone::v2	28	30	0	0	0
sentinelagent_alone_variant::v1	32	34	0	0	4
sentinelagent_alone_variant::v2	32	34	0	0	4
apc_sentinelagent_scope_only::v1	32	34	0	0	4
apc_sentinelagent_scope_only::v2	32	34	0	0	4
apc_sentinelagent_retained::v1	0	0	0	0	0
apc_sentinelagent_retained::v2	0	0	0	0	0
apc_sentinelagent_fail_closed::v1	0	0	20	28	0
apc_sentinelagent_fail_closed::v2	0	0	20	28	0

Reconciliation note (boundary, arm 2). The verbatim fields in `b\reconciliation.json` for item `arm2_canonical_leak_scope_overstated` are: `locked`: "boundary_condition.md and boundary_lock.json expected_per_arm_variant: sentinelagent_alone (arm 2, canonical) predicted 0 leaks / safe on all cases, for both v1 and v2"; `explanation` (excerpt): "The lock's 'arm 2 = 0 leaks / safe on all cases' overstated the prediction. … The variant-specific prediction holds: arm 2 safe on the variant cases; arm 5 leaks them; arm 2v leaks them standalone."; `resolution`: "recorded in reconciliation only; lock and artifacts unmodified; the variant-specific predictions (arm 2 safe on resource_widen / omitted_forbidden_default; arm 5 leaks them; arm 2v leaks identically standalone) all hold".

Table 11. Boundary seam counts per variant — the seam-test fields of the run manifest.

variant	arm 1 leaks (`arm1_leaks`)	seam cases (`seam_cases`)	seam count (`seam_count`)
v1	none	resource_widen-0-a, resource_widen-1-a, resource_widen-2-a, resource_widen-3-a	4
v2	none	omitted_forbidden_default-0-a, omitted_forbidden_default-1-a, omitted_forbidden_default-2-a, omitted_forbidden_default-3-a	4

Note: the manifest field `expected_vs_observed_ok = true` covers the seam count only (`seam_count == 4`).

4.5 Cross-cutting synthesis

The program's cross-cutting claim assembles the three pilots and the boundary run with the frozen status ⋆ `SUPPORTED`: under narrowing-preserving mappings, the seam carries less context than the receiving mechanism's checks would need to detect standing authority violations on their own, and composition creates no authorization failure that one of the mechanisms does not already exhibit standalone. The claim's conditions are part of the claim, and we state them with it. The mapping family is the first: the zero-seam result is a property of narrowing-preserving mappings (canonical T0; Section 3.2). The case domain is the second: it holds on the frozen case sets of the three pilots and the boundary run, and it is a recorded property of the cases built, not a theorem. The disclosed exceptions are the third: the E1 action family under arm-native scoring — measured and convention-dependent (Section 4.2, Table 7) — and the V1/V2 bridge artifacts, which exist identically standalone (Section 4.4, Tables 10–11). The recorded seam-zero readings sit in `tests\seam_pilot_v2\RESULTS_FOR_COAUTHOR.md` Section 1 (Pilot 1), in Section 3.3 of the extension's results, and in Section 1 of the SentinelAgent pair's results. The Pilot-1 zero is derived from the per-template identity of arm 2 and arm 5, not from an independently stored seam count.

The one measured composition-specific exception is the extension's action family, and its measurement is as much about scoring conventions as about the seam. Under the frozen family-wide norm, in which the mechanism-alone arm 2 is judged against the handoff-derived standard, the seam test counts zero; under the arm-native norm, in which arm 2 is judged against its own world's root authority, arm 2's re-scoring admits none of the 4 attack cases, the same records score 4 against the handoff norm, and the seam test counts 4 of the family's attack cases. The mini-run that made the receiver consume the narrowed action fact denied 4/4 attacks and allowed 4/4 benign calls with zero false denials (Section 4.2, Table 7); receipts verify 12/12, the arm-2 control parity and contract digests remain 8/8, and 10/10 locked predictions match (Table 7). We therefore report the exception as measured, convention-dependent, and bounded to one family with its 1 design point × 4 witness structure.

The boundary run supplies the companion attribution. A composed-only leak requires a non-narrowing-preserving

effective bridge and the receiver's decision to be the only thing standing between the widened call and execution; the boundary measurements discharge the derivation side of that proposition: the 4 new cases per variant leak in the composed arm and identically in the standalone variant arm, canonical arm 2 is safe on them, and the repair arms suppress them (Section 4.4, Table 10) — with the boundary's own effective sample size of 1 design point × 4 per variant. The forced identities — arm 2 ≡ arm 5 at 24/24 cases and 26/26 events in Pilot 1, arm 2 ≡ arm 5 at 28/28 in the SentinelAgent pair, and arm 2v ≡ arm 5 at 32/32 per variant in the boundary run — carry no evidential weight about composition; the informative contrasts, restated from the preceding subsections, are canonical arm 2 versus composed arm 5, and arm 2v versus arm 2. The construction-forced identities leave E1 as the only composition-specific test in the program; its result is bounded to one design point × four attack cases and is convention-dependent.

The formal core (Section 3.2) turns these observations into one account, with a different status vocabulary: the blindness lemma is definitional, its proposition measured clause by clause, with no frozen claim token. The lemma states that if the receiver's admission depends on the call only through its read-set projection (state included), and no value in that read set separates the narrowed set from its complement for a call pair, then no admission function over the read set can enforce the narrowing on that pair; its corollary is the program's aphorism, quoted verbatim: carried but unread = uncarried (for decision purposes). Its measured instances recur through this subsection with effective n attached: in the extension, the narrowed action set crossed the seam intact and the synthesis did not read it, and the consumption repair reversed the counts (Section 4.2, Table 7); in the SentinelAgent pair, the state/identity leaking classes are fields the receiver's vocabulary lacks, the witness set being 6 leak templates × 4 parameters = 24 attack cases of 48; the separate 4-case keyholder class is treated under H5/H6; in the boundary run, the widened manifest is admitted by the receiver standalone and composed — the bridge wrote the widened scope, so there is no consumption failure to repair at the receiver, and the derivation is the fault.

Those instances motivate the two repair loci. Receiver side, when the fact crossed intact: carry the upstream decision as evidence — signed attestation in the SentinelAgent pair, where the evidence arm eliminates all 28 leaks while recording no false denials itself; HMAC-signed M2-auth in the extension, where the scheme denies 4/4 attacks; digest-only evidence carriage in Pilot 1 — or retain the upstream denial, as the fail-closed arms do at their measured benign costs (Table 13) — with consumption, not carriage, as the part that enforces. Derivation side, when the bridge itself widened: fix the manifest derivation at the bridge, so that the derivation preserves the narrowing. Figure 2 places the two loci in one picture.

Table 12 collects the claim ladder as this section measures it: each row carries its frozen status where one is frozen, its locus, and the artifact evidence a reader can re-derive; the formal core, the extension's exception, and the boundary result are labeled measured.

Table 12. Claim-locus synthesis — frozen statuses, section loci, and re-derivable evidence.

claim / finding	status	locus	evidence (artifacts and cross-references)
H1 scope trust (action clause under M0; resource clause not supported; extended to the bridge's manifest derivation)	REFINED	Sections 4.1, 4.2, 4.4	`v2\raw_metrics.json` (`per_template.resource_widen.leaked`); `v2\extension_e1\metrics.json` (arm 2/5 leak fields); `dt\test1_rescored_arm2.json` (`counts`, `seam_test`); `b\metrics.json` (variant arms)
H2 dynamic state, identity, freshness leak; decision carriage closes	SUPPORTED	Sections 4.1, 4.3, 4.4	`v2\raw_metrics.json` (`tracegrant_alone`, `apc_tracegrant_retained`); `sa\sa\metrics.json` (`sentinelagent_alone`, `apc_sentinelagent_retained`); `b\metrics.json`
H3 state/identity vocabulary gaps manifest as silence, not misparse	SUPPORTED	Sections 4.1, 4.3, 4.5	`v2\raw_metrics.json` (`per_template.principal_rebind`, `per_template.composition_history_loss`, `per_template.repeated_calls`, `per_template.budget_reset`, `per_template.expired_delegation`, `per_template.revoked_envelope`); `sa\SA_RESULTS.md` Section 6; Section 3.2. Legacy H4 tags on recipient/default cases are not H3 evidence (Appendix Table A1).
H5 verification attests integrity and key possession, not authority	REFINED	Sections 4.3, 4.4	`sa\sa\controls.json`; `b\metrics.json`; `b\manifest.json`
H6 failure shapes; no new composition-specific failure mode except the disclosed E1 exception	SUPPORTED	Sections 4.1, 4.2, 4.3, 4.4, 4.5	`v2\raw_metrics.json`; `sa\sa\metrics.json`; `b\metrics.json`; `dt\REPORT.md` Sections 1–3
⋆ seam subtractive under narrowing-preserving mappings	SUPPORTED	Section 4.5	`sa\SA_RESULTS.md` Section 1; `v2\EXTENSION_E1_RESULTS.md` Section 3.3; `v2\claims_update.md`; `b\manifest.json` (`seam_test.v1.seam_count`)

claim / finding	status	locus	evidence (artifacts and cross-references)
Zero-seam conditionality (cross-cutting finding)	measured — no status token	Section 4.5	Section 3.2; `dt\REPORT.md`; `b\manifest.json` (`seam_test`)
E1 action-family exception (cross-cutting finding)	measured — no status token	Sections 4.2, 4.5	`dt\test1_rescored_arm2.json` (`counts.arm2_attack_leaks_handoff_norm`, `counts.arm2_attack_leaks_root_norm`, `seam_test`); `dt\test2_metrics.json`; `dt\REPORT.md` Section 3
Boundary bridge-layer attribution (cross-cutting finding)	measured — no status token	Sections 4.4, 4.5	`b\metrics.json` (`sentinelagent_alone_variant::*`, `apc_sentinelagent_scope_only::*`); `b\reconciliation.json` item `arm2_canonical_leak_scope_overstated`
Repair patterns (cross-cutting finding)	measured — no status token (instance claims carry their own statuses)	Sections 4.1, 4.2, 4.3, 4.4, 4.5, 5	Table 13; `dt\test2_metrics.json`; `sa\sa\metrics.json`; `v2\raw_metrics.json`
Keyholder finding (cross-cutting finding)	carried under H5 REFINED	Sections 4.3, 4.4	`sa\sa\controls.json`; `b\metrics.json`; `b\manifest.json`
Formal core (blindness lemma; seam proposition, both clauses measured; repair rule)	definitional lemma; measured proposition clauses — no status token	Sections 3.2, 4.5	definitions and lemma in Section 3.2; measured instances in Sections 4.2–4.4

Artifact-root abbreviations for Table 12 (display-only). `v2`, `dt`, `sa`, and `b` are defined in Section 3.5. Full paths remain in the artifact tree.

Legacy-evidence note. `v2\claims_update.md` is a legacy-key record. Its old-H6 block compares a local Task B fixture to Progent, is labeled directional-only, and is not used as evidence or as a baseline in this paper. The current H6 reading rests on the measured receiving pairs; Appendix Table A1 records the label lineage.

No unconditional reading is available from these runs, and each sentence above is bounded the way its evidence is: by its scoring convention, by its case set, and by the trust model of the arm it describes.

5. Discussion — when seams add authority; repair patterns

The practical question. This section answers the practical question the program was built to answer: when does a composed arm admit a call that no mechanism admits alone, and what repairs close that admission? The anatomy is the familiar one: a sender narrows authority, a bridge translates the narrowed decision plus state into a receiver-side record, and the receiver decides from that record. No party owns the seam. The measured answer is bounded: it holds for one family (the E1 action family), one scoring convention (arm-native), and the frozen case sets only. It is not a general claim about seams, and no production-safety claim is made for any repair arm. Every positive sentence below is conditional on the frozen mappings, case sets, and scoring conventions.

Where the composed arm alone admits. On the E1 action family, under arm-native scoring, the violation exists only in the composed arm. Arm 5 (M0 from the post-handoff snapshot) leaks `send_external` on 4/4 attacks, 4 unauthorized-effect events in total. Arm 1 (APC after the narrowed handoff) denies 4/4 with APC reason `action send_external not in scope`. Arm 6 (M2-auth signed evidence) denies 4/4 with `PROOF-GAP`, and all 8 signed records verify `EVIDENCE-VERIFIED`. Arm 7 (shadow denial retained) denies 4/4. Benign calls are allowed 4/4 in every arm, zero false denials. The attributable locus is the receiver's synthesis: arm 5's snapshot carries the narrowed action set `actions = [read_public, save_internal]` yet M0 never consumes the action field — carried but unread is uncarried for decision purposes (Section 3.2). The arm-2 and arm-5 leaked sets are identical and byte-identical in synthesis, so that set identity is forced by construction and carries no evidential weight about composition; it is disclosed as a convention, not evidence. This is precisely the blindness lemma's measured instance (Section 3.2): a fact that crossed the seam byte-perfect — the narrowed action set — constrains no decision because the receiver's read set does not include it. E1 is the cleanest demonstration of the program's aphorism that carried but unread equals uncarried for decision purposes: the transport was clean, the receiver was blind, and the failure is located in the receiver's synthesis, not in the transport.

Composition-specific on one family. Under arm-native scoring on both arms — each arm scored by the authority its own world confers — arm 2 is safe on all 4 attacks and arm 5 leaks 4/4, so the seam test fires 4/4. Under canonical scoring the same frozen records give 0/4 on this family; the difference is the scoring convention, and that dependence

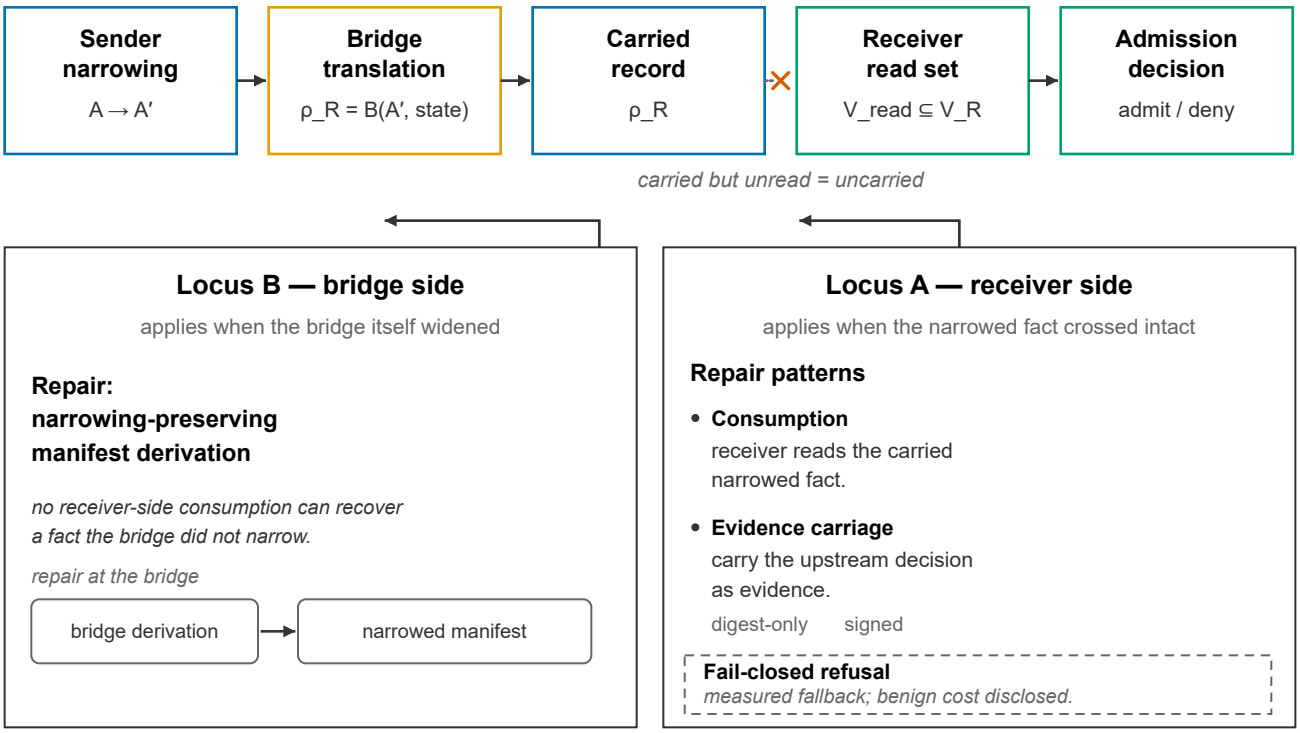


Figure 2. Where the decision can be rescued — the two repair loci of the seam program (schematic; no data). The upper band traces one delegated call through the pipeline these runs instrument: sender narrowing, bridge translation, the carried record, the receiver's read set, and the admission decision. Locus A sits at the receiver's admission step and applies when the narrowed fact crossed intact: consumption of the carried fact (the extension's mini-run) and evidence carriage of the upstream decision (digest-only in Pilot 1; signed in the SentinelAgent pair and E1 M2-auth), with the fail-closed refusal drawn as the measured fallback and its benign cost disclosed. Locus B sits at the bridge's derivation step and applies when the bridge itself widened: a narrowing-preserving manifest derivation, since no receiver-side consumption can recover a fact the bridge did not narrow. The annotation `carried but unread = uncarried` labels the gap between the carried record and the receiver's read set; Locus B points to the bridge-translation step, and Locus A points to that gap. Content is definitional-plus-measured (Section 3.2; the measured repair instances above); the figure carries no numbers and asserts no statuses.

is disclosed. Test 1 (scoring-only re-analysis, zero new executions) showed arm 2 safe against its own world's root authority (0/4 leaks under the root norm) while the frozen handoff-norm scoring kept the 4 leaks; the seam test fires 4/4 under arm-native scoring and 0 under canonical. Test 2 resolves attribution positively with execution: an M0 variant whose admission rule consumes `snapshot.actions` denies all 4 attacks with gate code `ACTION-SCOPE`, allows all 4 benign cases, verifies 12/12 receipts, and leaves the arm-2 control unchanged — 8/8 gate-outcome parity, 8/8 byte-identical contract digests, 10/10 locked predictions matched. Consumption, not carriage, is enforcement: the receiver must read the narrowed fact, not merely receive it. The `ACTION-SCOPE-ABSENT` fail-closed branch (missing or empty action field) was not exercised in this family because every snapshot carries a non-empty action set; it remains untested here.

Repairs. Four loci follow: three receiver-side (evidence carriage, retain the upstream denial, consumption) plus the derivation-side rule from Section 3.2 (Figure 2).

Evidence carriage (receiver). When the fact crossed intact, carrying the upstream decision as evidence closes the measured leaks. In the SentinelAgent pair, arm 6 (signed attestation) closes all 28 baseline leaks with 0 leaks and 0 false denials. In E1, arm 6 (M2-auth) denies 4/4 attacks; all 8 signed records verify `EVIDENCE-VERIFIED`. In Pilot 1, arm 6 carries the decision as a digest-only record and closes the 24/40 baseline leaks. The caveat is load-bearing: the signed arms assume the attestation is verified under the trusted key, and M2-auth narrows tamper and replay but does not establish authenticated transport — the signer is the same harness that was the original trust anchor, and any holder of the public test key can mint valid records. Evidence carriage is therefore a fix only relative to a

correctly deciding upstream: had the upstream APC admitted a widened scope, M2 would admit it too.

Retain the upstream denial (receiver). Keeping the sender's denial closes leaks at a measured utility cost. Pilot 1 arm 7 (fail-closed): 0 leaks on 40 attack cases, 20/40 benign false-denial cases, 28 false-denial events. SA arm 7: 0 leaks, 20 benign false-denial cases, 28 false-denial events. Boundary arm 7 (fail-closed): 0 leaks, SHADOW-DENY on the 4 seam cases per variant, 20 benign FD cases / 28 FD events. The utility cost is real: legitimate benign tasks are refused, and the cost is measured rather than asserted. Neither arm 6 nor arm 7 is production-safe; both are scoped design choices.

Consumption (receiver). Merely carrying the fact without reading it does not enforce. E1 Test 2: 4/4 attacks denied on `ACTION-SCOPE`, 4/4 benign allowed, 0 false denials, arm-2 control byte-identical — the narrowing-preserving meet's receiver-side half (Section 3.2). The general rule is to make the sender's boundary readable at the receiver and then read it: deny when the action snapshot is absent or empty (fail-closed), deny when the requested tool is outside the carried action set, otherwise run the unchanged synthesis. Evidence carriage is the transport; consumption is the enforcement.

Derivation-side narrowing-preserving meet (bridge). When the bridge itself widened the manifest (V1/V2), no receiver-side rule can recover a fact never narrowed. Boundary arm 5 leaks 32 cases / 34 events, which decompose as 28 pre-existing canonical leaks plus the 4 new cases per variant ($32 = 28 + 4$); standalone arm 2v leaks the identical 4 per variant; canonical arm 2 is safe on the variant-specific cases; arms 6/7 suppress APC-DENY / SHADOW-DENY $\times 4$ per variant; seam count 4 per variant. Composition neither creates nor removes the failure: the identical admissions appear standalone in the variant arm. The fault is the bridge's manifest derivation — V1 widens role scope beyond root R, V2 default-fills the SEND endpoint — so the repair belongs at the bridge, not the receiver. Fix the derivation at the bridge.

Keyholder finding (H5). Signature verification attests key possession and payload integrity, not who minted the record; any keyholder is equivalent to the issuer within the token vocabulary. This holds for delegate-minted tokens — `keyholder_widen` admitted 4/4 in arms 2/5 — and for the bridge's own manifest derivation (V1/V2). SA signature controls: 4 `SIGNATURE_INVALID` per token arm for foreign-key probes; all 10 controls passed. Key distribution is the seam's critical interface: integrity plus key possession is not source of authority, so trusting a signature is not the same as trusting the minter.

Positioning. This paper positions two deliverables: a reproducible conformance suite designed for others to run against their own bridge/mechanism pair before deploying, and a tightened design rule — narrowing preservation at the mapping layer — for bridges that are not narrowing-preserving. The kit leads: it packages the frozen case sets, the harness, and the locked-prediction protocol so other teams can run the same seam tests against their own bridge/mechanism pair. The design rule survives in tightened form: adopt narrowing preservation at the mapping layer, with the measured receiver-locus repair (consumption) applied when the fact crossed intact and the bridge-level fix (derivation) applied when it did not. A broader expansion is a post-workshop decision, never a commitment made here. A kit reports; it does not fix. No roadmap; no production-safety claim.

Statuses. H1 `REFINED`; H2 `SUPPORTED`; H3 `SUPPORTED`; H5 `REFINED`; H6 `SUPPORTED`; $\star$ `SUPPORTED`.

Disclosures. Zero seam holds only for narrowing-preserving mappings (canonical T0) on the frozen case sets, with the exception and the bridge artifacts disclosed. E1's exception is convention-dependent: family-wide vs arm-native scoring; one family; 1 design point $\times$ 4; widening-only operationalization caveat; the seam test as operationalized can fire only where the composition widens. Arm 6 assumes attestation verified under the trusted key; arm 7 accepts 20 benign false denials; neither arm is production-safe; M2-auth does not establish authenticated transport. No claims beyond the frozen sets.

Table 13 collects the repair loci with their measured results.

Table 13. Repair patterns — four repair loci with measured results.

Repair locus	Repair	Measured result
(a) receiver-side evidence carriage	carry upstream decision as evidence (digest-only in Pilot 1; signed in SA and E1 M2-auth)	Pilot 1 arm 6: 0 leaks (24/40 baseline closed), 0 false denials; SA arm 6: 0 leaks (all 28 baseline leaks closed), 0 false denials; E1 arm 6 M2-auth: 4/4 denied
(b) receiver-side retain the upstream denial	retain upstream denial (fail-closed / shadow deny)	Pilot 1 arm 7: 0 leaks; 20 benign FD cases; 28 FD events; SA arm 7: 0 leaks; 20 benign FD cases / 28 FD events; boundary arm 7: 0 leaks; SHADOW-DENY $\times 4$ per variant; 20 benign FD / 28 FD events
(c) receiver-side consumption	consume narrowed action fact before admission	E1 Test 2: 4/4 attacks denied `ACTION-SCOPE`; 4/4 benign allowed; 12/12 receipts; arm-2 parity 8/8, digests 8/8; 10/10 predictions

Repair locus	Repair	Measured result
derivation-side rule	narrowing-preserving meet at bridge	Boundary arm 5: 32 cases / 34 events = 28 pre-existing + 4 new per variant (32 = 28 + 4); standalone arm 2v leaks identical 4 per variant; seam count 4 per variant

6. Related work

Purpose. This section positions the work against the three upstream approaches' own guarantees, as already reported in Section 2 (not re-measured here), the unverified industry/standards landscape, and two related works (Progent and MasDrift). It reports no new results; guard language applies to others' numbers throughout. There are no tables and no figures in this section.

Upstream formal spines. APC's "Composition Soundness" (Thm 4.7) is intra-session composition of action sequences, conditional on a complete effective restriction set, serialized admission, and static X — a within-mechanism guarantee, not a cross-mechanism composition guarantee. TraceGrant's soundness is assumed (Eq. 10), not proven. SentinelAgent's Thm 4 is composition safety for two chains that both satisfy P1–P7 within its own framework; it does not cover a chain handed to a different mechanism's decision procedure. None of the three covers cross-mechanism composition as this program defines it: sender narrows, bridge translates, receiver decides, with no owner at the seam. APC's meet operates on scope within a single mechanism and is conditional on static X; TraceGrant's soundness is assumed rather than proven and does not extend to a cross-mechanism handoff; SentinelAgent's theorem covers two chains within its own framework, not a cross-mechanism handoff. Each guarantee stops at its own boundary. This paper's narrowing-preservation rule plus the blindness lemma (Section 3.2) covers the slot the three guarantees leave uncovered: the moment authority crosses from one mechanism's vocabulary into another's. No upstream mechanism is claimed broken or failing; all upstream figures remain their authors' self-reports, and "reports" is the operative word, never "achieves".

Industry / standards landscape. These are pointers only; their bibliographic identities were checked, but their claims are unverified and none is treated as a verified fact. Noma (Apr 2026) is cited as an adjacent industry framing of inter-agent trust at handoffs; the IETF draft (revised Aug 2026) is cited as a related cross-domain authorization-sharing proposal. Neither is a measurement of the specific cross-mechanism translation gap this program instruments, and neither is adopted here as a result or a fix. arXiv 2605.05440 is cited as a related reference in the same area. None of these is a benchmark, a reproduced result, or a verified input to this manuscript; each citation is a pointer whose claims were not reproduced, and the frozen claim set governs if any pointer differs from it.

Progent (reference only). Progent is cited as a close related work on monotonic confinement inside a single privilege-control layer. No Progent figure is reproduced, re-measured, or compared here; the paper uses no Progent number as evidence, and no Progent result is treated as an input or a baseline. The word is "reports," never "achieves"; Progent is cited as a reference only.

MasDrift (related work). MasDrift studies authorization preservation across multi-agent coordination architectures using benign productivity tasks [MasDrift]. Our narrower question concerns executions through two specific public authorization artifacts and their argument-binding paths; the small deterministic matrix is a conformance study, not an architecture benchmark and not a substitute for MasDrift's broader evaluation. No MasDrift number is reproduced, re-measured, or compared.

Disclosures. Others' numbers stay self-reported. Nothing claims APC, TraceGrant, SentinelAgent, Progent, or MasDrift is broken or fails; no vulnerability in any upstream mechanism as published is claimed. No production-safety claim; no benchmark comparability. No claims in this section go beyond the frozen sets. The single industry/standards pointer set (Noma, the IETF draft, arXiv 2605.05440) is described as pointers, not verified facts.

Scope of positioning. This section positions the paper against the three measured formal spines and two related works (Progent and MasDrift); it does not survey the broader delegation and capability literature (token exchange, macaroons, relationship-based authorization, capability systems). No novelty claim over that literature is made, and none of those systems is treated as a baseline; a comparative treatment would require additional verified references and is not undertaken here.

7. Threats to validity

This section states what the measured program does not claim: the construct limits of the case sets, the internal limits of the reconciliation record, and the external limits of interpretation. Two hypotheses remain open as future instruments; every other limitation is a plain disclosure, not a softened claim. Table 14 maps each threat to the section that raises it.

7.1 Construct threats

The case sets are constructed witnesses, not benchmarks. The first pilot family is 10 templates × 8 runs and the SentinelAgent family is 12 templates × 8 runs, each written for this program and frozen before any run. Their structure, not their scale, is what licenses a reading: the E1 extension rests on a single design point with a neutralized parameter axis, measured as 1 design point × 4 attack cases, and the boundary run rests on two design points, each 1 × 4 cases — 8 attack cases in total, which are not 8 independent cases.

Convention dependence is a construct threat as much as a disclosure. The E1 seam test scores 0 under the frozen family-wide convention and 4/4 under the arm-native convention; the convention was locked and disclosed, not chosen after the fact. SentinelAgent scoring is arm-invariant, and rescoping it would flip `composition_history_loss` arm 2 from leak to safe. At the boundary, arm 2 is always canonical regardless of the variant loop variable while the variant arms use the variant manifest derivation, so the identity of arm 2v with arm 5 per variant is forced by construction. The composed arm's absolute count is accordingly $32 = 28 + 4$ — the pre-existing canonical set plus the per-variant new cases. Three forced identities — Pilot 1 arm 2 ≡ arm 5 at 24/24 cases and 26/26 events, SentinelAgent arm 2 ≡ arm 5 at 28/28, and boundary arm 2v ≡ arm 5 at 32/32 per variant — carry no evidential weight about composition and are labeled forced.

Two operational caveats complete the construct picture. As operationalized, the E1 seam test can fire only where the composition widens: a zero where it does not does not by itself establish composition-independence, and the same caveat may attach to the original pilots' seam test, which was not re-verified on that point. Separately, arm 2's authority assignment in E1 is a disclosed convention — the own-world norm is the one discriminating Test 1 adopts, and under the alternative reading the same records locate the failure in the mapping's authority assignment, still not in a mechanism interaction.

7.2 Internal threats

The reconciliation record is part of the evidence and was never edited. Within the SentinelAgent reconciliation, the two checks that failed are recorded verbatim: false-denial events locked as 26 and observed as 28, and the arm-5 confusion true-negative denominator locked as 68 and observed as 20 on the attack-case basis. Both carry `affects_behavior_cells: false` in that record, which is why they change no outcome cell and remain on file. At the boundary, the reconciliation records `expected_vs_observed_ok` as a scope check — it verifies the manifest's expected seam count and nothing beyond it — and the canonical arm-2 leak scope was carried as overstated: 0 in the locked reading against 28 pre-existing observed cases.

Two constraints limit the repair results from inside. Arm 7's shadow-denial action rule (`tool not in snapshot.actions`) was not independently exercised: the retained shadow denial fired first in all four E1 attacks. And M2-auth does not establish authenticated transport — the key is a public test constant, the signer is the same harness that was the original trust anchor, and there is no forger control. On the trust side, arm 6 assumes the attestation is verified under the trusted key and arm 7 accepts 20 benign false denials; neither repair arm is claimed production-safe, and neither should be read as one.

7.3 External threats

V1/V2 are plausible bridge designs; they may be mis-designs, and they are not asserted to be typical, the only, or the most likely bridges. No generalization beyond the frozen case sets is claimed. No vulnerability is claimed in APC, TraceGrant, or SentinelAgent as published: the measured findings are properties of this implementation under the locked bridge mappings and key models, and the upstream mechanisms' own evaluations remain their authors' self-reports; this manuscript reports those self-reports and assesses none of those systems. No comparability with published benchmark numbers is claimed, neither theirs against this program's counts nor this program's against theirs. Finally, the zero-seam statement — composed arms adding no admission beyond the receiver's own, under

narrowing-preserving mappings (canonical T0) on the frozen case sets, with the E1 exception and the bridge artifacts disclosed as measured — has the standing of a recorded property of every case built, not a theorem.

7.4 The not-claimed canon

The frozen not-claimed list is restated here by group, without softening and without upgrading.

Scope of the experiments. Not claimed: bridges beyond V1/V2 — the two variants are this program's own constructed designs, and typicality is not asserted. Not claimed: generalization beyond the frozen case sets, whose witness sets are as stated in Section 7.1. Not claimed: comparability with published benchmark numbers, since the case sets are constructed here and upstream evaluations are self-reported by their authors. Not claimed: that the seam-count result generalizes beyond these pilots' case families; it is a recorded property of every case built, not a theorem.

Upstream mechanisms. Not claimed: any vulnerability in APC, TraceGrant, or SentinelAgent as published; the findings are properties of this implementation under the locked bridge mappings and key models.

Repair patterns. Not claimed: production safety of arms 6 or 7 — arm 6 assumes the attestation is verified under the trusted key, and arm 7 accepts 20 benign false denials; both are scoped design choices. Not claimed: authenticated transport for M2-auth — the key is a public test constant, the signer is the same harness, and no forger control exists. Not claimed: arm 7's shadow-denial action rule was independently exercised — it was not, because the retained shadow denial fired first.

Conventions. Not claimed: convention-invariance of the seam test — the counts differ under frozen family-wide scoring versus arm-native scoring (0 versus 4/4), and the canonical disclosure is unchanged and remains accurate. Not claimed: rescoping-neutrality of SentinelAgent scoring, which is arm-invariant and would flip `composition_history_loss` arm 2 from leak to safe under rescoping. Not claimed: boundary-convention invariance — arm 2 is always canonical regardless of the variant loop variable, and arm 2v matches arm 5 per variant by construction. Not claimed: that the seam test as operationalized measures more than where the composition widens; a zero where it does not would not by itself establish composition-independence. Not claimed: an unchallenged authority assignment for arm 2 in E1 — the own-world norm is a disclosed convention. Not claimed: that H3 covers caught representable cases, the E1 action-family consumption failure, or the V1/V2 derivation widening; those are reported under H1/H6 and the measured findings, not H3.

Open hypotheses. Not claimed here: the shared-audit-trail hypothesis and the intent-layer hypothesis — carried below as unbuilt instruments, not as findings and not as claims.

7.5 Two hypotheses carried open

H4 — **`UNTESTED.`** This is an unmeasured surface, not a demonstrated failure: H4's claim is that there is no shared audit trail across the mechanisms — the SHA-256 evidence chain on the sender-mechanism side, the receiver's receipts, and the hash-linked tokens are three separate provenance records, and this hypothesis is not tested in any pilot. The SentinelAgent tokens in the runs are one-hop, each parent hash resolving to the root, so no chain reconstruction and no cross-mechanism trail interchange was exercised in either the SentinelAgent pilot or the boundary run. The next instrument is identified but not built here: a case family that reconstructs authority from a token hash chain and compares it with the APC envelope's history. Nothing adjacent is imported into this claim.

H7 — **`UNTESTED.`** The intent layer of the SentinelAgent pair, its principle P2, may be gameable through benign-word override, and this was not tested: P2's `intent_text` was carried in the run records but never classified, and neither the E1 extension nor the boundary run exercised it. No probing instrument is introduced in this paper, and the hypothesis stands as open.

Table 14. Threat ledger — each threat and the section that raises it.

Threat	Raised in
Bridges beyond V1/V2 out of scope; V1/V2 not asserted typical	Section 7.3
Production safety of the repair arms not claimed	Sections 7.2, 7.4
Shared audit trail and intent layer carried open, not claimed	Section 7.5
Generalization beyond the frozen case sets not claimed	Sections 7.1, 7.4
No vulnerability claim about the upstream mechanisms as published	Sections 7.3, 7.4
Comparability with published benchmark numbers not claimed	Sections 7.3, 7.4
The seam-count result is a recorded property, not a theorem	Sections 7.3, 7.4

Threat	Raised in
Family-wide versus arm-native convention dependence	Sections 7.1, 7.4
SentinelAgent arm-invariant scoring; rescoping flip	Sections 7.1, 7.4
Boundary arm-2 canonical convention; forced arm-2v identity	Sections 7.1, 7.4
M2-auth does not establish authenticated transport	Sections 7.2, 7.4
Arm 7's shadow-denial action rule not independently exercised	Sections 7.2, 7.4
E1 seam test operationalized only where the composition widens	Sections 7.1, 7.4
Arm-2 authority assignment as a disclosed convention	Sections 7.1, 7.4

8. Reproducibility and artifacts

This section describes the conformance kit prepared for release — the program's practical deliverable at workshop stage. It is protocol description only: what was frozen, what was locked, and what was re-verified. It carries no result claims.

8.1 What the kit contains and how it was run

The kit packages the four frozen experiment sets — the two pair pilots, the action-clause extension with its resolving tests and controls, and the boundary run — as runnable case sets with their harnesses under the locked-prediction protocol. Predictions, cases, arms, and locked input hashes were frozen before execution; the one recorded harness-hash deviation is Test 2's runner hash, disclosed in Section 4.2; executions were strictly sequential, offline, and deterministic; discrepancies between predicted and observed values were reconciled verbatim into files that were never edited after the fact; and each published number was generated from the artifacts and embedded byte-verbatim. The execution ledger records the scale: 2,446 = 560 + 40 + 6 + 16 + 672 + 1,152.

Re-execution discipline is part of the kit. The SentinelAgent program recorded determinism reruns of 20 executions with 0 mismatches, and its control battery passed 10/10; the boundary run's control battery passed 7/7. The controls exist so that a receiver-side reader can check claims about repair results before trusting them. Scoping limits are carried with the kit as well: there is a single implementation per repair pattern, so each repair measurement is scoped to its own implementation; the kit reports what a mapping carries and does not fix anything; and sanitization and cleanup of the release tree remain pending. The kit is a release candidate, not a public release: repository URL, license, DOI, and archival location are not yet assigned.

8.2 The artifact hash ledger

Table 15 lists six frozen addresses; two are also quoted in the text, and the table is not a complete hash inventory. The numbers in this paper are traceable through the artifact tree to the artifacts that generated them. Table 15 transcribes the frozen addresses for the E1 dataset, the discriminating-tests lock, the SentinelAgent lock and case set, and the boundary lock and harness, in each case from the frozen source map that records each value and its artifact root.

Table 15. Artifact hash ledger — frozen lock/case/harness hashes.

Artifact	Frozen hash
E1 dataset (canonical `dataset_sha256`)	`1b7c6083…b2e264ba`
Discriminating lock (file SHA-256)	`8cd4d993…996011ab`
SA lock (file SHA-256)	`9f2611a1…65d47f0`
SA cases (file SHA-256)	`ca2f05d5…e42cb6d2`
Boundary lock (file SHA-256)	`b4deab1a…082c4296`
Boundary harness (file SHA-256)	`fa8f7e42…790814af4`

Note: row 1 is the canonical dataset hash recorded in the E1 lock; rows 2–6 are file SHA-256 values. The table is a citation aid, not a complete hash inventory.

Suffix note. Displayed hashes are variable-length truncations; row 3 uses a 7-character suffix and row 6 uses a 9-character suffix. Full 64-character values are in the ancillary `hashes.txt`.

9. Conclusion

This paper asked a narrow question about a broad practice: when one authorization mechanism narrows a decision and hands it to another through a bridge, what crosses, and what can the receiver no longer derive? The question matters because the composed pair's decision is what executes, while each mechanism's own guarantee — conditional on its own envelope, its own contracts, its own static assumptions — stops at the handoff that nobody owns.

The method was chosen to make the answer auditable rather than merely reported. Predictions, case sets, arms, and hashes were locked before any run; executions were sequential, offline, and deterministic; mismatches were reconciled verbatim; and every published number was generated from the artifacts. Four experiment sets put 2,446 = 560 + 40 + 6 + 16 + 672 + 1,152 executions behind the measured statements below.

The measured answer is conditional, and its conditions are part of the answer. Under narrowing-preserving mappings (canonical T0), on the frozen case sets, the seam is zero: composition adds no admission beyond what the receiving mechanism exhibits standalone. A single composition-specific exception was measured, in the E1 action family under arm-native scoring, and it is convention-scored — the same records score 0 under the family-wide convention. The exception is one design point × four attack cases and is not generalizable beyond the frozen family. Two of the three pair-level zeroes are construction-forced because the composed and standalone receiving arms share the same contract and scoring; E1 is the only genuine composition-specific test, and it is convention-dependent. A receiver-side repair was measured alongside it: when the receiver consumes the narrowed fact rather than merely carrying it, the admitting calls stop; carriage and consumption are different things; receiver-side enforcement requires consumption, not carriage alone. At the boundary, two non-narrowing-preserving bridge designs produced failures that appear identically in the standalone variant arm — a bridge-design artifact, not a composition-specific failure, with composition neither creating nor removing the fault. And the keyholder caveat stands across all of this: a signature verifies integrity and key possession, not the source of authority, so anyone holding the key can mint records the receiver cannot distinguish from the issuer's; the key distribution is where this seam's trust actually lives. Every positive sentence here is conditioned on the frozen sets, the locked mappings, and the scoring conventions under which the counts were produced.

The practical deliverable is the conformance kit of Section 8: frozen case sets, harnesses, and the locked-prediction protocol, packaged so that others can run these seam tests against their own bridge and mechanism pairs before deploying one. A kit reports; it does not fix.

What remains open is stated in Section 7 as unbuilt instruments rather than findings: no shared audit trail has been measured across the mechanisms, and the sentinel pair's intent layer has not been probed. The closing status of the measured program is: H1 `REFINED`, H2 `SUPPORTED`, H3 `SUPPORTED`, H5 `REFINED`, H6 `SUPPORTED`, and ⋆ `SUPPORTED` for the cross-cutting zero-under-preservation finding — with the two open hypotheses above carried forward as instruments, not claims — H4 `UNTESTED`, H7 `UNTESTED`.

Continuation beyond the workshop decision point is not a commitment made here; any expansion is a post-workshop decision taken on its own evidence.

Appendix A. Hypothesis lineage and operationalization

The frozen experiment artifacts predate the current hypothesis key, and some labels changed meaning between the pilot records and this manuscript. Table A1 records the lineage so that a legacy label is not read as the current label with the same number. It is provenance, not evidence, and it changes no status.

Table A1. Hypothesis lineage — legacy artifact labels, their operationalizations, and the current manuscript labels.

Legacy artifact label	Legacy operationalization	Current manuscript label	Status / provenance
H1 (all artifact sets)	Receiving mechanism re-derives scope or trusts the already-authorized caller	H1	`REFINED`; unchanged label
H2 (all artifact sets)	Dynamic state, identity, and freshness do not cross and leak	H2	`SUPPORTED`; unchanged label
H3 (v2 only)	VERIFY postconditions are an admission/completion boundary	No direct current-H3 equivalent; historical completion-integrity finding	Reported under H1/H6 and the measured findings, not current H3. `v2\claims_update.md` H3; `raw_metrics.json` `unsatisfiable_verify`

Legacy artifact label	Legacy operationalization	Current manuscript label	Status / provenance
H4 (v2 only)	Mapping choices decide whether a failure is a mechanism property or an artifact	No direct current-H4 equivalent; historical mapping-attribution finding	Reported under H1 `REFINED` and the measured findings, not current H4. `case_generator.py` tags recipient/default cases old H4; those are caught-case evidence, not current H3 or H4
H6 (v2 only)	Comparison with the Progent reference numbers	No current-H6 equivalent; current H6 is similar-pair vs dissimilar-pair failure shapes	Historical, directional-only Progent block; not used as a comparison or baseline. Current H6 rests on the SA/boundary measurements
H3–H6 (SA/boundary)	Vocabulary mismatch; no shared audit trail; verification gap; similar/dissimilar failure shapes	H3–H6	H3 `SUPPORTED` (narrowed to state/identity gaps); H4 `UNTESTED`; H5 `REFINED`; H6 `SUPPORTED`
H7 (manuscript only)	Intent layer P2	H7	`UNTESTED`; `intent_text` carried but never classified

The frozen artifacts are not edited; this table is the reconciliation. In particular, recipient/default cases tagged old H4 in `tests\seam_pilot_v2\design\case_generator.py` are caught-case evidence and are not current H3 or current H4 evidence.

Path notation. The roots `v2`, `dt`, `sa`, and `b` are defined in Section 3.5; full paths remain in the artifact tree.

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Reference status: APC, TraceGrant, SentinelAgent, Progent, and MasDrift are arXiv preprints; no peer-reviewed venue is asserted. Noma is an industry blog; the IETF entry is an active individual Internet-Draft, not an RFC. Bibliographic identities were checked on 20 October 2026 (seven entries) and 22 October 2026 (MasDrift); the works were not reproduced, and none is treated as a verified input to the measurements.